

Algorithm Selection meets Bayesian Optimization for Tabular Regression

Bharadhwaj Manchikanti, Reddy Sai Tapasvy Kasaba Sai, Jennifer Sreedhara Jeeyar
Hyper-Optimists

Modality 1/3

Motivation

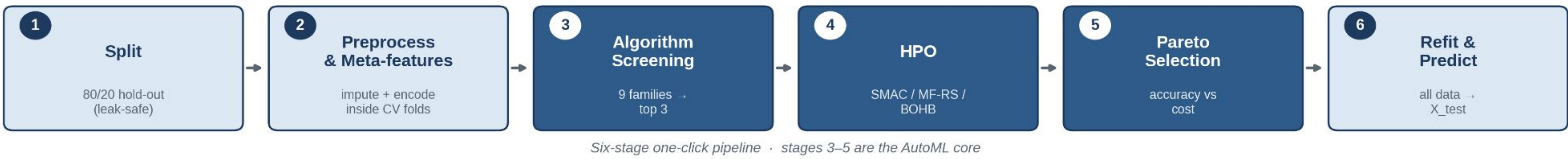
Performance on tabular regression tasks is strongly dataset-dependent. A model family or hyperparameter configuration that works well on one dataset may perform poorly on another and exhaustively tuning every candidate is computationally expensive.

Therefore, we developed a modular **AutoML pipeline**: a reproducible, resource-aware AutoML system that automatically identifies a strong model and configuration for previously unseen tabular regression datasets.

Contributions

- Jennifer Sreedhara Jeeyar:**
Preprocessing, model and search-space design, meta features extraction, pipeline design
- Bharadhwaj Manchikanti:**
Algorithm screening, compared multiple HPO strategies, including Bayesian and Multi-Fidelity Hyperparameter Search
- Reddy Sai Tapasvy Kasaba Sai:**
End-to-end AutoML pipeline integration, pareto analysis, experimental evaluation and reproducibility

Our Approach



METHODOLOGY:

- **Split** - Creates an 80/20 development split for training and validation
- **Preprocessing** - Median impute + One-hot, fitted inside every CV fold so no fold sees the others
- **Meta-features** - 9 descriptors (size, skew, scale, correlation) that characterise each dataset
- **Algorithm Screening** - Quick-fits all 9 model families on 30% of data; only the top 3 advance to HPO, saving budget
- **Hyperparameter Optimization** - SMAC (random-forest surrogate + expected improvement) tunes the survivors with k-fold CV; multi-fidelity RS and BOHB are compared alongside it in the ablation
- **Pareto Analysis** - keeps non-dominated trials that trade off accuracy against training time
- **Refit + Predict** - Rebuilds the selected pipeline on all training data and generates predictions for X_{test}

OBSERVATIONS:

- We ran a controlled ablation - same models, same budget, only the optimiser changes : [Defaults](#), [Random Search](#), [Multi-fidelity RS](#), [SMAC \(Bayesian Opt\)](#) and [BOHB](#), over 3 seeds.
- SMAC ranked best overall (avg rank 2.0 of 5) and was the most stable across seeds. HPO beat tuned defaults on 4/5 datasets - the gain is largest on hard, unsaturated data (wine +0.09) and negligible where defaults already saturate (brazilian). BOHB added cost without accuracy: data-fraction fidelity saves too little on fast tree models to pay off.
- On the exam dataset SMAC reached 0.926 honest hold-out and scored $R^2 = 0.935$ on the hidden test - beating our own estimate. Full run: 47 min, 3.3% of the 24 h budget.
- TabPFN foundation model - competitive on small data, but CPU 1000-row limit rules it out at our scale.

Exam test score: $R^2 = 0.9351$

- Week 1
- Week 2
- Week 3
- Week 4
- Week 5
- Week 6
- Week 7
- Week 8
- Week 9
- Week 10
- Bonus
- Literature

Resources Used

For development:
- CPU intel core i5 12th gen
- CPU intel core i5 8th gen
- macOS

For AutoML:
- CPU intel core i5 12th & 8th gen
- Runtime < 1 hour of permitted 24 hours

Workforce:
- 1.5 full week on average

Number of queries for test score generation: 1

Empirical Results

